Learning Objective 3.3: I can identify the hardware architecture of the ADALM-PHASER and control it via SDR software.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Walking the receive chain.** A wave from an X-band source arrives at the PHASER's patch face and ends up as numbers in the Raspberry Pi.

   (a) List, in order, the blocks the signal passes through from a single patch to the Raspberry Pi. Name the part number or device for each one.

   (b) The ADAR1000 sets element phase with a 7-bit phase shifter. Compute the phase LSB in degrees, and state what physically limits how finely you can steer the beam.

   LSB = $\boxed{}$

   (c) How many array elements feed one ADAR1000, and how many RF channels leave the pair of ADAR1000s on their way to the mixers?

   elements/chip = $\boxed{}$

   RF channels = $\boxed{}$

   (d) The element spacing is $d = 14$ mm. For a source at 10.525 GHz, find the wavelength and the spacing in wavelengths.

   $\lambda =$ $\boxed{}$

   $d/\lambda =$ $\boxed{}$

2. **The frequency plan.** The PHASER mixes the received X-band signal down to a fixed 2.2 GHz IF using high-side injection, so $f_{\text{LO}} = f_{\text{RF}} + f_{\text{IF}}$. The ADF4159 and HMC735 VCO cover 12.2 to 13.0 GHz. The Pluto is tuned to 2.2 GHz and sampled at 3 MSPS.

   (a) **Find HB100** reports the lab source at 10.42 GHz. What LO frequency must the software command?

   $f_{\text{LO}} =$ ☐

   (b) Given only the VCO limits and the fixed IF, what range of source frequencies can this receiver reach?

   $f_{\text{RF,min}} =$ ☐

   $f_{\text{RF,max}} =$ ☐

   (c) A student skips **Find HB100**, so the software keeps the nominal LO of 12.725 GHz. The source in front of the array is actually at 10.30 GHz. What IF does the mixer produce, and by how much does it miss 2.2 GHz?

   $f_{\text{IF}} =$ ☐

   $\Delta f =$ ☐

   (d) Find the width of the baseband window the Pluto digitizes at 3 MSPS, and say in one or two sentences what the student of part (c) sees on the FFT tab.

   window = ☐

3. **Hybrid beamforming.** The PHASER has eight elements, eight RF phase shifters, and two receive channels into the SDR.

   (a) Why does the board put a phase shifter behind every element at RF instead of digitizing all eight elements and doing the phase shift in software?

   (b) After the two 4:1 analog sums, how many independent complex weights can software still apply, and what information has been destroyed?

   digital weights =

   (c) An adaptive nulling algorithm running on the digital channels can steer roughly one null per available degree of freedom. State one capability the hybrid split keeps and one it takes away, and say which of the two you would trade if the board's job were to null several jammers at once.

   (d) How many ADC channels would a fully digital 8-element version need, and name one practical cost besides part count.

   ADC channels =

4. **Reading the bench.** Lab Preset 1 is loaded, the HB100 is at boresight about 1 m from the array, and the FFT tab is showing the baseband spectrum. The Pluto is tuned to 2.2 GHz and the LO is at 12.725 GHz.

   (a) The FFT peak sits 1 MHz above the center of the window. Find the IF of the received tone and the frequency of the source.

   $f_{\mathrm{IF}} =$ $\boxed{}$

   $f_{\mathrm{RF}} =$ $\boxed{}$

   (b) **Rx Gain** is raised from 10 dB to 30 dB. Predict what happens to the peak level, to the noise floor, and to the separation between them, and explain why.

   (c) A student reports no peak anywhere in the FFT. Give the first two checks you would make, in order, and say what each one rules out.

   (d) On a uniform-taper beam sweep this board normally shows a noise floor about 23 dB below the peak. A student's sweep shows only 8 dB. Give two plausible causes and say how you would tell them apart.

Documentation: ____